Student Number:									

Surname:																
First Name(s):																

Department of Mathematics and Applied Mathematics

MAM2000W: 2LA (Linear Algebra)

July exam **6 July 2021**

Time : 2 hours **Maximum marks: 60**

Full marks: 60

- This question paper consists of 14 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space. Give an indication in your answer when you are switching to the back of a page.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators are allowed.
- For Section A, only final answers will be marked. For Section B, full answers must be given.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
7	7	4	14	14	8	6	**60**

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Section A

In this section, give your final answer only. Do not show your working.

Question 1. [7 points] The matrices B and U are given below. The matrix A is the matrix B with its last column deleted.

$$B = \begin{pmatrix} 2 & 0 & 1 & 3 & -1 & a \\ -4 & -1 & 0 & -6 & 2 & b \\ 0 & 3 & -6 & 3 & 1 & c \\ 2 & -1 & 3 & 6 & 0 & d \end{pmatrix} \qquad U = \begin{pmatrix} 2 & 0 & 1 & 3 & -1 & a \\ 0 & -1 & 2 & 0 & 0 & b+2a \\ 0 & 0 & 0 & 3 & 1 & c+3b+6a \\ 0 & 0 & 0 & 0 & 0 & d-9a-4b-c \end{pmatrix}$$

The matrix U is a row-echelon form of B (you need not check this) and $\mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ 3 \\ k \end{pmatrix}$, where k is a constant.

(a) For which value(s) of k will the system of linear equations $A\mathbf{x} = \mathbf{b}$ have (i) no solutions (ii) a unique solution (iii) infinitely many solutions?

(i) No solution:

(ii) Unique solution:

(iii) Infinitely many solutions:

(b) Give the rank and nullity of A.

Rank(A) = $\qquad$ Nullity(A) =

(c) (i) Give a basis for $NS(A)$.

(ii) Give a basis for $CS(A)$.

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Question 2. [7 points] You are given that $T : P_2(x) \to M_{2\times 2}(\mathbb{R})$ is a linear transformation and that

$$T(1) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad T(x) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad T(x^2) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

(a) Find $T(ax^2 + bx + c)$.

(b) Give a basis for the range of T.

(c) Give the kernel of T and a basis for it.

(d) Find the matrix A representing T with respect to the ordered bases

$$B = \{1, x, x^2\} \text{ and } C = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

for $P_2(x)$ and $M_{2\times 2}(\mathbb{R})$ respectively.

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Question 3. **[4 points]** For every $p(x), q(x) \in P_2(x)$, put $\langle p(x), q(x)\rangle = \int_0^1 p(x)q(x)\,dx$. You may assume that this defines an inner product on $P_2(x)$. Let $r(x) = x^2$

(**a**) Let $p(x) = ax^2 + bx + c$. Find $\langle r(x), p(x)\rangle$ in terms of a, b and c.

(**b**) Find $||r(x)||$.

(**c**) The subset A of P_2 is defined by $A = \{ax^2+bx+c \in P_2(x) : (ax^2+bx+c) \perp r(x)\}$. An incomplete description of A is given below; there should be an equation in the place of the dots ....

$$A = \{ax^2 + bx + c \in P_2(x) : \ \ldots\}.$$

Write down this equation.

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Section B

In this section, give full answers to all the questions. Show all your working.

Question 4. [14 points] Let $S = \{A \in M_{2\times 2}(\mathbb{R}) : \det(A - A^T) = 0\}$ and

$$B = \left\{ \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \right\}.$$

Prove or disprove each of the following statements:

(a) S is closed under scalar multiplication.

(b) S is a subspace of $M_{2\times 2}(\mathbb{R})$.

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(c) $\begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \in \operatorname{span}(B)$.

(d) B is linearly independent.

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(**e**) B is a basis for $M_{2\times 2}(\mathbb{R})$.

(**f**) B is a basis for $\operatorname{span}(B)$.

(**g**) $B \cup \left\{ \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \right\}$ is a basis for $M_{2\times 2}(\mathbb{R})$.

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Question 5. [14 points] Let $A = \begin{pmatrix} 4 & 0 & 2 \\ 0 & -5 & 0 \\ -4 & 0 & -2 \end{pmatrix}$.

(**a**) Find all the eigenvalues of A.

(**b**) Give reasons for your answers to the following two questions, using only your answer to (a).

(**i**) Can A be written as a product of elementary matrices?

(**ii**) Can A be diagonalized?

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(**c**) For each eigenvalue, find a basis for the corresponding eigenspace.

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(**d**) Find, if possible, matrices P and D such that D is diagonal and $P^{-1}AP = D$.

(**e**) Find an elementary matrix A and an upper triangular matrix T such that $A = ET$.

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Question 6. [**8 points**] Read the following proof and then answer the questions following it.

Let A be a square matrix and $\lambda_1, \lambda_2, \lambda_3$ be distinct eigenvalues of A with corresponding eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

It can be shown that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent. (1)
Suppose that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent.
Then there are scalars μ_1, μ_2, μ_3, not all zero, such that

$$\mu_1\mathbf{v}_1 + \mu_2\mathbf{v}_2 + \mu_3\mathbf{v}_3 = \mathbf{0} \tag{2}$$

Then $\quad A(\mu_1\mathbf{v}_1 + \mu_2\mathbf{v}_2 + \mu_3\mathbf{v}_3) = A\mathbf{0}$ (3)
and hence

$$\mu_1\lambda_1\mathbf{v}_1 + \mu_2\lambda_2\mathbf{v}_2 + \mu_3\lambda_3\mathbf{v}_3 = \mathbf{0} \tag{4}$$

Multiply (2) by λ_3 and subtract (4) from it to obtain

$$\mu_1(\lambda_3 - \lambda_1)\mathbf{v}_1 + \mu_2(\lambda_3 - \lambda_2)\mathbf{v}_2 = \mathbf{0} \tag{5}$$

From this it follows that $\quad \mu_1(\lambda_3 - \lambda_1) = \mu_2(\lambda_3 - \lambda_2) = 0,$ (6)
and hence $\quad \mu_1 = \mu_2 = 0.$ (7)
Substituting (7) in (2) yields $\mu_3\mathbf{v}_3 = 0$, and hence $\mu_3 = 0$. (8)
But this is a contradiction, (9)
and hence $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ must be linearly independent.

(**a**) Explain how the equation (4) follows from equation (3).

(**b**) What is used to deduce (6) from (5)?

(**c**) Where in the proof is the fact that the eigenvalues are distinct used, and how is it used?

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(d) Why can we say at (8) that if $\mu_3 \mathbf{v}_3 = 0$, then $\mu_3 = 0$?

(e) What is the contradiction referred to at (9)?

(f) Prove the statement at (1). [Hint: Use a proof by contradiction.]

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Question 7. [6 points] A square matrix B is called *nilpotent* if there is a positive integer k such that B^k is the zero matrix.

(a) Give an example of a non-zero 2×2 nilpotent matrix.

(b) Prove that if B is nilpotent, then 0 is an eigenvalue of B, and in fact the only eigenvalue.

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(**c**) Prove that if a 2×2 matrix has characteristic equation $\lambda^2 = 0$, then it is nilpotent.

(**d**) Prove or disprove: A diagonalizable square matrix with a non-zero eigenvalue cannot be nilpotent.